

Project Plan: A.C.E (AI Casino Education)

Background: There are a lot of people that want to gamble in casinos but get overwhelmed by the amount of strategies and probability that comes into play. A.C.E. will create a fun and interactive way to play and learn more about casino games and help users improve their skills to improve their chances of winning more money.

Objectives:

- Develop a working and interactive AI
- Create a working database that accurately reflects data
- Make a fun and interactive UI
- Integration with different parts perform adequately

Project Outline: This project is a full-stack web application using a Python/Flask backend, a React-based frontend, and PostgreSQL for data storage, divided into four independently developed layers: game logic, UI design, AI advisory, and database management. The application launches from a single entry point and presents a main menu with a series of buttons. Playing the game will start a count of a virtual chip count that gets saved to the database upon exit. Players can then easily access the database to view their net chip balance and win to loss record.

Deliverables: The final deliverables for this project will include a live hosted web application that users can access through a browser. The web app will have the following features: a complete Blackjack gameplay loop with working logic to simulate a real game and a tutorial mode with an AI trainer that teaches the user the best moves to make for any given hand. Both modes will include virtual chip managers to keep track of the player's balance, while also ensuring each game starts with the same, static balance. Players will also be able to access a Stats page through a database created using PostgreSQL, which will store net chip balances, showing how much in total the player has made or lost, incrementing after each completed game. Secondary deliverables will include the full source code uploaded to a GitHub repository, a README with setup and deployment instructions, a test suite for all game logic and API endpoints. Stretch goals, such as a Texas Hold 'Em Poker and Roulette game will be implemented if core features are functional by Week 10.

Evaluation Criteria:

The success of the project will be evaluated based on unit tests covering any and all hand scenarios and payout calculations, all of which must pass. Strategy accuracy will be verified by cross-referencing the AI advisor with public strategy charts across over 20 different hand scenarios. UI usability will be assessed by having two players unfamiliar with the rules of

Blackjack play a round without any assistance. Database reliability will be tested after each session to confirm the necessary stats are getting updated accordingly.

Project Plan and Timing:

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